

FERRITIC/AUSTENITIC TRANSITION WELD + NDE PROPOSAL

A dissimilar transition weld between **ferritic steel** and **austenitic stainless steel** is considered for a high pressure corrosive system.

WELDING

MATERIAL PROPERTIES

Ferritic: needs to withstand high pressure forces (**strength**) and absorb lots of energy before fracture (**toughness**)

Austenite: needs to withstand highly corrosive environments (**corrosion resistance**)

Weld filler: should be at least **equal in strength and corrosion properties** to the poorest component in the joint

MATERIAL SELECTION

Weld is between a **low carbon alloy steel (SA508)** and an **austenitic stainless steel (305)** using an austenitic **weld filler (309)**

Material	C	Si	Mn	Ni	Cr	Mo	Nb	Cr Equiv	Ni Equiv
SA 508	0.19	0.1	0.25	3.4	1.6	0.45	0.01	2	9
Type 305	0.05	0.25	1.0	10	19.0	0.05	0.05	19	12
Type 309	0.01	0.65	1.0	12	25.0	0.25	0.01	26	13

CHEMICAL COMPOSITION

Carbon: Precipitation and solid solution strengthening.

Silicon: Used for steelmaking purposes; killing the steel and removing oxygen. i.e improves resistance to oxidation. (important for high integrity)

Manganese: Improves strength and hardness

Chromium & Nickel: Primarily to drive corrosion resistance, but nickel can also improve toughness and hardenability.

Molybdenum: Enhances creep resistance

Niobium: Grain refinement and precipitation hardening

NICKEL AND CHROMIUM EQUIVALENTS

The **nickel equivalent** and **chromium equivalent** differ from the nickel content and chromium content, respectively.

$$\text{Cr Equiv.} = \text{Cr} + \text{Mo} + 1.5\text{Si} + 0.5\text{Ni}$$

$$\text{Ni Equiv.} = \text{Ni} + 30\text{C} + 0.5\text{Mn}$$

Higher chromium equivalent will promote ferritic microstructure as its composed of **ferritic stabilisers**, likewise higher nickel equivalent will promote austenitic structure as its composed of **austenite stabilisers**.

WELDING CONSIDERATIONS

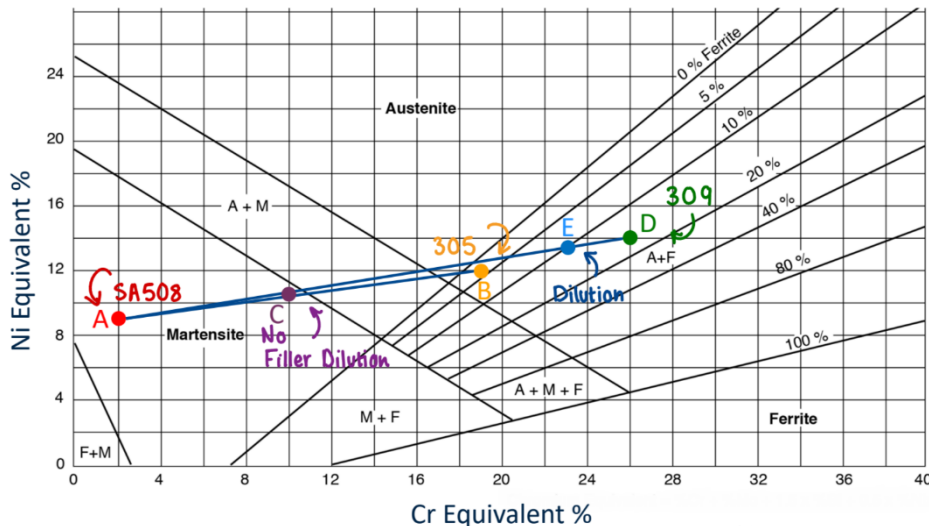
Dilution control: the resultant chemical composition of the fusion zone is related to the degree of mixing between the filler metal and the two parent metals.

Sensitisation: where carbides form from chromium, and essentially deplete the chromium content.

Unwanted phases: if metal is unstable, delta ferrite and sigma ferrite phases can form and degrade to brittle intermetallic phases, becoming susceptible to corrosion and fracture.

Ferrite Number (FN): A low ferrite number suggests larger proportion of austenite and hence better corrosion resistance, while a high ferrite number infers higher cracking resistance. **Ideal FN for dilution of weld filler in this case = 10**, which gives **good balance of crack resistance and corrosion properties**.

SCHAEFFLER DIAGRAM



Point A: SA508 Ferritic

Point B: 305 Austenite

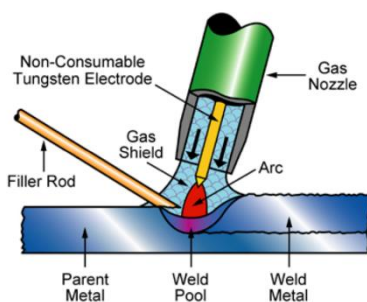
Point C: Welded area if no filler weld was used

Point D: 309 Filler

Point E: Target composition (FN=10) of weld controlled by dilution factor

Position (and hence composition) of E also depends on the size of the weld. A wider weld means a larger dilution factor and hence larger displacement on Schaeffler diagram. See below for more info on the geometric weld design:

WELDING PROCESS



The weld volume in this context is very small with **steep fusion angles of about 4 degrees**, resulting in very tight joint gaps. Hence there isn't much of a dilution factor.

TIG (Tungsten Inert Gas) performs well in this kind of weld design as it is well suited for narrow joints and steep angles with fine control of heat input. Inert gas also prevents oxidation

NON-DESTRUCTIVE EVALUATION (NDE)

NDE is comprised of seven main areas, two of which (highlighted) are the main focus for this study: **Flaw detection and evaluation** and **Location determination and evaluation**.

NDE METHODS: RADIOGRAPHIC VS ULTRASONIC TESTING

NDE method needs to be a volumetric inspection method that is able to both locate and size the defects; the choice is between **radiographic testing (RT)** and **ultrasonic testing (UT)**.

UT is very versatile it is able to size and locate defects, whereas RT don't offer the same resolution in depth due to the nature of radiography and hence locating defects is more difficult using RT.

TYPES OF ULTRASONIC TESTING:

Longitudinal waves: produced by voltage applied to a piezo-electric transducer.

Transverse (shear) waves: require more acoustically solid material for effective propagation (not effective in liquids and gases).

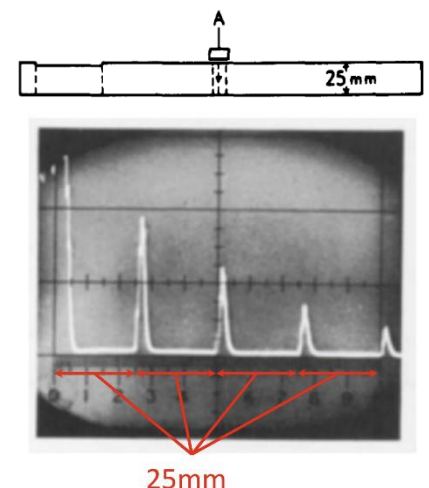
Longitudinal waves are approx. twice the speed of shear waves, hence wavelength (resolution) of transverse waves is roughly half, making it more sensitive to smaller defects.

However, **propagation of transverse waves is more difficult through anisotropic structure of weld**. So ultimately, sizing tolerance is hindered because longitudinal waves (which suffer from worse resolution due to larger wavelength) are more effective.

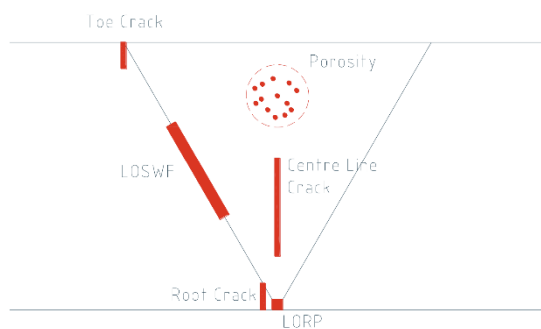
HOW DOES UT WORK:

Signals representing interface reflection are produced: Initial bang followed by evenly spaced repeating backwalls (intervals depending on component thickness) of decreasing amplitude.

If there is a defect, the waves will reflect in between these fixed intervals. This will appear as a smaller sharper **new echo (peak)** before the first backwall echo depending on the defect size/orientation.



DEFECT TYPES & DETECTION METHOD:



Toe Crack: two techniques could be used:

- Transverse waves from a 45° probe
- Creep waves (78° longitudinal surface waves)

LOSWF: best response given by 60° transverse probe which will strike defect at 90°

LORP: 45°, 60° or 45° transverse probes from either side of centreline

Root Crack: similar to LORP, but will not produce defined rise/fall signals and instead will produce multiple peaks and troughs.

Porosity: Difficult with UT as signals produced would be very weak, RT can be more effective

Centreline Crack: uses a transmitting shear probe and a receiving shear probe in a tandem setup.

Defect Type	Detection Method
Toe Crack	UT; transverse waves at 45°, or creep waves
LOSWF	UT; transverse waves at 60°
LORP	UT; transverse probes preferred at 45°
Root Crack	UT; transverse probes preferred at 45°
Porosity	Difficult with UT; RT can be more effective
Centreline Crack	UT; using two transverse probes

MISCELLANEOUS – QUESTIONS & CONSIDERATIONS

How do we filter out more esoteric alloys in material selection?

Exotic materials can have excellent properties which are far in excess of what's required -- most popular ones tend to be the cheaper ones, which we can manufacture in bulk. They need to be produce to a large scale and sufficient volume, hence cost becomes a factor that cannot be ignored

Material index considerations

We need to maximise both fracture toughness and yield strength, not necessarily a ratio between them. Density is not an important factor for this context.